Small Hotel Prototype Review

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On July 30, 2026, the Summary of Major Comments section was updated to follow the structure of four groups that later reviews contained. Changes shown in square brackets [].

Small Hotel Scorecard Spreadsheet

Summary of Major Comments

The following comments represent errors found when comparing values from the scorecard compared to the source cited in the scorecard:

- Wall U-value for 1978-1997 vintage (SHSC30-Envelope)
- Roof U-value for 2019+ vintage (SHSC31-Envelope)
- Window Guest Room U-factor for pre 1978 and 1978-1997 vintages formula error (SHSC34-Envelope)
- Infiltration values incorrect (SHSC41-Envelope)
- PTAC EER too low (SHSC46-HVAC System)
- PVAV cooling efficiency too low (SHSC47-HVAC System)
- No exhaust needed for stairways and storage rooms (SHSC56-Ventilation)
- Water heater set point not consistent with reference (SHSC60-Water Heater)
- Elevator load 1 kW low (SHSC74-Equipment)

The following are the highest priority comments:

- Too many people assumed in the lobby (SHSC27-Zones) [moved from other important issues]
- Guestroom occupancy is insufficiently diverse (SHSC28-Zones) [moved from other important issues]
- Gas heating for PTAC may not be possible (SHSC43-HVAC System, SHSC44-HVAC System) [moved from other important issues]
- Guestroom thermostat settings are insufficiently diverse (SHSC53-Thermostat) [moved from other important issues]

- Guestroom ventilation rate too low (SHSC54-Ventilation) [moved from other important issues]
- Guest room occupancy schedules are insufficiently diverse (SHSC80-Schedules) [moved from other important issues]
- Guest room lighting power density is high (SHSC64-Interior Lights) [moved from other important issues]
- Assembly Occupancy schedule overused for lobby, gym, office, laundry, and rest rooms (SHSC76-Schedules) [moved from other important issues]

The following comments represent other important issues found:

- No shading from entrance overhang (SHSC10-Prototype)
- Wood framed roof with steel framed walls (SHSC11-Prototype)
- Missing indoor swimming pool (SHSC19-Zones)
- Roof replacement for pre 1978 and 1978-1997 vintages should not use Title 24-1992 (SHSC32-Envelope)
- SHGC for 1998-2007 should use non-North and North values (SHSC35-Envelope, SHSC36-Envelope)
- Fan motor and drive efficiency should vary by size (SHSC50-HVAC System)
- Water heater sizing not sensitive to water main temperature (SHSC61-Water Heater)
- Lobby interior lighting LPD and plug load inconsistent with large hotel (SHSC62-Interior Lights, SHSC70-Equipment)
- Lobby HVAC availability schedule should be 24x7 (SHSC78-Schedules)

In addition, the following comments describe the lack of disclosed assumptions that make verifying the values in the scorecard nearly impossible:

- None

Background

The review of the “score card,” which is a spreadsheet summarizing the model assumptions, is broken down by spreadsheet tab, which includes:

- Notes
- Prototype
- Zones
- Envelope
- HVAC System

- Thermostat
- Ventilation
- Water Heater
- Interior Lighting
- Ext Lighting
- Equipment
- Schedules
- Energy Usage
- Weights
- PV
- EV Charger

The comments on these tabs are labeled using the following format SHSCx-<TAB>-<CELLRANGE>), where <TAB> is the name of the tab in the spreadsheet and <CELLRANGE> indicates the

SHSC1-Notes (n/a)

Sometimes, the specific benefits of code changes are not captured in a set of prototype models because they do not include a building with a specific feature. This has become an issue for PNNL, where a great fraction of the addenda for 90.1 with energy savings cannot be quantified using the models. See Table 4 of PNNL's "ANSI/ASHRAE/IES Standard 90.1-2022: Energy Savings Analysis - February 2024," where many addenda with energy savings were not included in the quantitative analysis and instead were just discussed qualitatively. 23 of the 39 were not captured quantitatively. In some ways, general prototypes cause this exclusion of understanding impacts, rather than custom prototypes that look at a particular measure. I'm not suggesting a change, but it is a warning that a good system of prototypes still does not answer all questions. Note this is the same comment as PPD1 in my single-family residential prototype review and LHSC1-Notes in the large hotel review.

SHSC2-Notes (n/a)

The impact weight of a measure can be misleading when using a standardized set of prototype models. The scenario is when a measure is part of a technology that is represented in only one or a few prototype buildings in the set. In that case, the impact of the measure may be more indicative of the weight of that building rather than the weight of the technology in general. Take air-cooled chillers, for example. Perhaps they appear in only

one prototype model for commercial buildings. An improvement measure for air-cooled chilled water plants could be significant, but the weight of one building may show that it has a small impact. This is because the weight of one building is based on the building type weighting, which may have nothing to do with the weighting of the air-cooled chiller market. In fact, it would be impossible to weigh the market size for every technology across a standardized set of building prototypes since the data usually does not exist. This subtlety makes it important to understand the frequency of appearance of a given technology in the set of prototypes, which is very important to disclose when assessing the overall impact of a measure. Note this is the same comment as PPD2 in my single-family residential prototype review and LHSC2-Notes in the large hotel review.

SHSC3-Notes (A1:C21)

Additional data sources could be added to the list shown in the spreadsheet:

- American Hotel and Lodging Association (AHLA) 2018 Lodging Survey https://www.ahla.com/sites/default/files/2018%20Lodging%20Survey%20Final.pdf
- HVS U.S. Hotel Development Cost Survey 2024 https://www.hvs.com/article/9999-HVS-US-Hotel-Development-Cost-Survey-2024
- Northwest Energy Efficiency Alliance – Commercial Building Stock Assessment 2019 https://neea.org/commercial-building-stock-assessment/
- ENERGY STAR Score for Hotels in the United States https://web.archive.org/web/20240513175940/https://www.energystar.gov/sites/default/files/tools/Hotel.pdf
- Energy Efficiency Indicators for Hotel Buildings https://www.mdpi.com/2071-1050/13/4/1754
- Natural Resources Canada – Survey of Commercial and Institutional Energy Use 2019 https://oee.nrcan.gc.ca/corporate/statistics/neud/dpa/menus/scieu/2019/tables.cfm
- ASHRAE Advanced Energy Design Guide for Highway Lodging https://www.ashrae.org/technical-resources/aedgs/30-percent-aedg-free-download
- Technical Support Document – The Development of the Advanced Energy Design Guide for Highway Lodging Buildings – PNNL 17875 https://www.osti.gov/biblio/939043

- Technical Support Document – 50% Energy Savings Design Technology Packages for Highway Lodging Buildings – PNNL 18773 https://www.pnnl.gov/main/publications/external/technical_reports/pnnl-18773.pdf

By utilizing additional sources of information, the validity of assumptions can be strengthened. Some parameters may be California-specific, but comparing assumptions across multiple databases lends credibility to any assumptions made, even when a California-specific database ultimately may be the source of the information. Tables showing parameters across multiple databases, and then values chosen being highlighted, can be a powerful expression of when parameters are solidly determined by being similar in multiple data sets versus being based on just a single data set. Please also note that some of the above references may include citations of relevant reports and papers that could be reviewed as further sources. Note this is the same comment as PPD2 in my single-family residential prototype review and LHSC3-Notes in the large hotel review.

SHSC4-Notes (B2:B23)

Many of the abbreviations being used on the data sources are familiar to building energy analysts but may not be as familiar to others. Please add an unabbreviated version of each item, as well as providing links as a reference (missing for the first four).

SHSC5-Notes (B22:C22)

The link for the PNNL 20405 report does not point to a location that is accessible. It looks like a local file server. Instead, it should point to:

https://www.pnnl.gov/main/publications/external/technical_reports/pnnl-20405.pdf

SHSC6-Notes (A40:J58)

The EIA data pointing toward a large number of lodging being under 10,000 sqft and 30 percent of the buildings being one floor points to the large number of existing single-floor motels that often had fewer than a few dozen rooms. While this represents a large number of buildings, it does represent very much of the constructed small hotels in the last 50 years, which have been primarily “highway hotels” with a double-loaded corridor and usually three stories. I agree that using a three-story “highway hotel” as the type of building as the prototype makes sense, with a floor area of about 30,000 sqft. It would be interesting

to see the average age of the small hotels in each floor area bin shown by the EIA data to see if this is substantiated. Perhaps more justification on this tab may help others understand the decision on the size and number of floors better.

SHSC7-Notes (A62:D72)

The use of NUMPRO=1 is not explained. I think it means “Number of Projects,” but that is not explained at all. The phrase “all extended” is also used without explanation. Also, the columns labeled “Total” next to the lower two tables are not explained and are kind of confusing as to what is being totaled.

SHSC8-Notes (n/a)

The hospitality industry seems to categorize hotels into

- Luxury
- Upscale
- Midscale
- Economy
- Independent

Given that the room size, amenities, type of occupancy, and other features may be different for these different categories, it would be good for the prototype to be put into one of these categories. Another set of groupings that the hospitality industry uses is:

- Urban
- Suburban
- Airport
- Interstate
- Resort
- Small Metro/Town

And the prototype should probably be described as one of these categories. In addition, if the properties that would impact energy use vary greatly across these groups, additional prototype variations may be warranted. Note this is the same comment as LHSC6-Notes in the large hotel review.

SHSC9-Prototype (D6)

The document link for “NonRes-Vintage bins-Draft Report-30Nov2023-Final.docx” is to a SharePoint that may not be broadly accessible.

SHSC10-Prototype (A13:C13)

The model shows no shading geometry; this is very unlikely for the lobby entrance, which almost always has an overhang over the entrance to protect visitors from rain when checking in or checking out of the hotel.

SHSC11-Prototype (A11:C11)

The use of a wood-framed roof may be inconsistent with the use of steel-framed walls.

SHSC12-Prototype (A12:C12)

It would be good to describe the average window area, height, and width for the guest rooms, or perhaps this should be included as additional columns on the Zones tabs for all zones.

SHSC13-Prototype (n/a)

It would be good to document how many guest rooms the hotel has and the average size of each. I believe 57 rooms based on the Zones tab, with five on the first floor, 26 on the second floor, and 26 on the top floor, averaging 330 sqft in size. This is consistent for a typical room in a highway hotel. The number of rooms on the first floor seems low.

SHSC14-Prototype (n/a)

Based on the Zones tab, the fraction of the total area used by guest rooms is 58%. This seems reasonable for this type of hotel, although still a little lower than I would have expected. The CPUC large hotel was 64%. Please show this figure on the Prototype tab.

SHSC15-Prototype (n/a)

More information was provided on the large hotel scorecard related to the existing prototypes and how this prototype is compared. It would be good to include that type of information on this tab.

SHSC16-Prototype (A7:C7, A12:C12)

The orientation of the building is not indicated, but given that the WWRs are described as north and south with the largest fractions, I presume the image is showing one of those directions. It would be good to clarify which direction is facing north to understand the solar impact on the building.

SHSC17-Prototype (A14:C15)

The floor-to-floor and floor-to-ceiling heights are shown as the same. While a common modeling simplification, it is not very realistic.

SHSC18-Zones (n/a)

Please add a table that summarizes the total area and fraction of area by the space type.

SHSC19-Zones (n/a)

This size hotel often has an indoor swimming pool for guests. It is very unusual for highway hotels or other three-story hotels not to have a swimming pool. Natatoriums are a highly specialized type of zone and have a disproportionately large impact on the overall building energy use. In addition, energy codes often have special rules for heaters and pumps related to pools that could be captured if included in the building. While outdoor pools are common in climates that are always warm, in many climates where a good portion of the year would not support having an outdoor pool, an indoor pool is often present.

SHSC20-Zones (n/a)

Please include the floor-by-floor plan views for the building for each unique floor. Please also identify the zone types by color.

SHSC21-Zones (A9:J9)

The closed office size of 1404 ft2 seems large for this size hotel. I think the front desk (which I presume is actually part of one of the two Lobby+Lounge zones) is not included, which means the office is just the office for the hotel manager and staff. This is seldom bigger than a typical office for a single individual. Seven people are shown as using this office. That is much larger than expected. Even during busy check-in times, it is unusual for a hotel this size to have more than two or three staff working at the front desk and office.

SHSC22-Zones (n/a)

Hotels this size typically have a small kitchen that provides service for breakfast and sometimes evening snacks. The kitchens are very small with seldom much more than a sink, microwave, and refrigerator.

SHSC23-Zones (n/a)

Hotels this size typically have a computer room available to guests, usually the size of a small office with one or two computers and a printer.

SHSC24-Zones (n/a)

Hotels this size typically have a small guest laundry room with a pay-per-load washer and dryer. It is possible this is part of the Laundry_FloorG zone, but it is not clear, and a staff laundry room for cleaning linens would probably operate on a very different schedule.

SHSC25-Zones (n/a)

Hotels this size typically do not have spas or other luxury facilities, so the absence of them from the zone list makes sense.

SHSC26-Zones (A6:J8)

Why are two different Lobby + Lounge zones defined?

SHSC27-Zones (A6:J8)

The number of people in the two Lobby + Lounge zones is 70, which seems very high. The hotel has only 56 rooms and 56 room occupants, which would mean that almost everyone is expected to be in the Lobby + Lounge zones at the same time, and some people would have guests. That seems extremely unlikely.

SHSC28-Zones (A16:J39)

For guest rooms, the number of occupants is shown as 1, but this is just a rounded value, and it is generally 1.1 people per guest room based on 3 Persons/1000 ft2 in column J, and when showing more digits in column I. Given the large fraction of the building that guest rooms represent, it is important to model them accurately. It might make more sense to try to derive an estimate of occupancy, based on literature suggested in SHSC3-Notes or from other sources, to find how often a room has no one, one person, a couple, three, or four people occupy hotel rooms and create a schedule that mixes up those integer values of people so that the overall occupancy matches the statistics. The use of lighting, equipment and water in rooms with zero to will clearly be different than one or four people. Note this is essentially the same comment as LHSC36-Zones for the large hotel prototype.

SHSC29-Envelope (A3:X17)

The opaque values seem to be identical to the large hotel scorecard for:

- External Wall_Guest Room
- External Wall_Nonres
- Roof

So the issues found when reviewing that are repeated below as SHSC30-Envelope, SHSC31-Envelope, and SHSC32-Envelope.

SHSC30-Envelope (M4, M9)

The exterior wall guest room and nonresidential for 1978-1997 vintage is shown as 0.256, which is based on Rt=3.9, but the value from Title 24-1986 shows Rt=4.6, which should be a U-value of 0.217. This column value was probably copied from the CZ6 value by accident. This is a repeat of the comment for the large hotel LHSC24-Envelope.

SHSC31-Envelope (K17, M17)

The roof U-Value for 2019+ is based on T24-2025, and the note says, “The lowest U-factor between Nonres and Guest room was considered,” but the values for CZ3 and CZ5 are incorrect. The U-Value shown for both CZ3 and CZ5 is 0.034, but the Nonres value from Table 140.3-B for “wood framed and other” shows 0.28 for both CZ3 and CZ5, which is lower. The 0.34 value is probably from Table 140.3-C for a hotel. This is a repeat of the comment for the large hotel LHSC25-Envelope.

SHSC32-Envelope (A13:X14)

The roof values are based on Title 24-1992, as explained in the note: “Considering 30 years for roof/ insulation replacement. (Guest room requirement)” and the values are correct, but the presumption that T24-1992 is a good stand-in for this in 2025 seems incorrect. If one assumes that all these roofs were replaced 30 years ago, then Title 24-1995 would make more sense. Since the odds of all the roofs being replaced in 30 years are slight, a better assumption would be:

- The average value of roof requirements from Title 24 from 1995, 1998, 2001, 2005, 2008, 2010, 2013, 2016, 2019, 2022, or
- The value of roof requirements from 15 years ago, so using Title 24-2010 (less accurate but still justifiable).

This is a repeat of the comment for the large hotel LHSC26-Envelope.

SHSC33-Envelope (many)

Many of the fenestration values seem to be identical to the large hotel scorecard for the following rows:

- Windows_ Guest Room
    - U-Factor 1998-2007 (row 21)
    - U-Factor 2008-2018 (row 22)
    - U-Factor 2019+ (row 23)
    - SHGC 2008-2018 (row 26)
    - SHGC 2019+ (row 27)
- Windows_ Nonres
    - U-Factor 1998-2007 (row 31)
    - U-Factor 2008-2018 (row 32)

    - U-Factor 2019+ (row 33)
    - SHGC 2008-2018 (row 36)
    - SHGC 2019+ (row 37)

The rows that were checked are:

- Windows_ Guest Room
    - U-Factor pre 1978 (row 18) – error found, see comment SHSC34-Envelope
    - U-Factor 1978-1987 (row 19) – error found, see comment SHSC34-Envelope
    - SHGC pre 1978 (row 23) - ok
    - SHGC 1978-1987 (row 24) – ok
    - SHGC 1998-2007 (row 25) – error found, see comment SHSC35-Envelope
- Windows_ Nonres
    - U-Factor pre 1978 (row 28) – error found, see comment SHSC34-Envelope
    - U-Factor 1978-1987 (row 29) – error found, see comment SHSC34-Envelope
    - SHGC pre 1978 (row 33) - ok
    - SHGC 1978-1987 (row 34) - ok
    - SHGC 1998-2007 (row 35) – error found, see comment SHSC36-Envelope

SHSC34-Envelope (I18:X19, I28:X29)

Based on the description “ASHRAE Handbook of Fundamentals (Table 4-Ch15) for single pane & T24, 2013 for double pane (33%-67%)”, the formula:

=0.338*1.12+0.67*0.36

Should be

=0.33*1.12+0.67*0.36

I’m not sure I understand where the 8 after the 0.33 came from.

SHSC35-Envelope (A25:X25)

For row 25, for the guestroom SHGC for 1998-2007 using Title 24-2001, the values shown on the table are all the North values instead of the non-North values for the 11%-20% range. These should be fixed.

	Non-North	North	Non-North	North	Non-North	North	Non-North	North	Non-North	North
0-10% WWR	0.46	0.68	0.41	0.61	0.47	0.61	0.36	0.49	0.36	0.47
11-20% WWR	0.46	0.68	0.40	0.61	0.40	0.61	0.36	0.49	0.31	0.43
21-30% WWR	0.36	0.47	0.31	0.61	0.36	0.61	0.31	0.40	0.26	0.43
31-40% WWR	0.30	0.47	0.26	0.55	0.31	0.61	0.26	0.40	0.26	0.31

In addition, cell H25 calls these “10-20%” instead of “11-20%” as it appears in the table.

SHSC36-Envelope (A35:X35)

For row 35, for the nonresidential SHGC, the values shown on the table are all the North values instead of the non-North values for the 11%-20% range. These should be fixed. For Climate zones 6, 7, 8, and 9, the values are correct, but only because the north and non-north are the same requirement.

	Non-North	North	Non-North	North	Non-North	North	Non-North	North	Non-North	North
0-10% WWR	0.49	0.72	0.61	0.61	0.61	0.61	0.47	0.61	0.46	0.61
11-20% WWR	0.43	0.49	0.55	0.61	0.61	0.61	0.36	0.51	0.36	0.51
21-30% WWR	0.43	0.47	0.41	0.61	0.39	0.61	0.36	0.47	0.36	0.47
31-40% WWR	0.43	0.47	0.41	0.61	0.34	0.61	0.31	0.47	0.31	0.40

In addition, cell H25 calls these “10-20%” instead of “11-20%” as it appears in the table.

SHSC37-Envelope (A25:X25, A35:X35)

For the SHGC for 1998-2007, the Title 24-2001 version was used, but that version has different SHGC values for north and non-north directions; however, the values from the table are for non-north only. The prototype for this version of the standard should probably use the north values for all fenestration facing north. This is a repeat of the comment for the large hotel LHSC28-Envelope.

SHSC38-Envelope (all)

Instead of using a single version of Title 24 to represent a range, perhaps it would make sense to use an average value for all versions of Title 24 within the time period. In other words, for the 1978-1997 range, values from the 1978, 1980, 1982, 1984, 1986, 1987, 1988, 1992, and 1995 versions of Title 24 should all be extracted and averaged together. Currently, from 1978-1997, uses values from Title 24-1992 only. For the time range 1998-2007, which could average the values from the Title 24 versions from 1998, 2001, and 2005

instead of using just the 2001 values. For the time range 2008-2018, we would average the values from Title 24 versions from 2008, 2010, 2013, and 2016 instead of using just the 2013 values. This is a repeat of the comment for the large hotel LHSC27-Envelope.

SHSC39-Envelope (A18:X19, A23:X24, A28:X29, A33:X34)

Averaging the U-Factor and SHGC between single and double-pane options leads to non-physically possible windows. The weighting between single and double pane is based on a statistic that showed 33% of small hotels that were built before 2000 were single pane from CBECS 2018. The weighted value is shown correctly, but the concept of ending up with a type of fenestration that is not physically possible may cause problems with simulation engines such as EnergyPlus that expect a window to be built up from a specific pane of glass with specific properties. It might be better to have a mixture of single-pane and double-pane windows in a model (although that is a different compromise with reality). This is almost an exact repeat of the comment for the large hotel LHSC29-Envelope.

SHSC40-Envelope (many)

Assuming only fixed windows and no operable windows may be appropriate for larger hotels in some climates, but operable windows are common in many hotels in temperate climates. This is a repeat of the comment for the large hotel LHSC30-Envelope.

SHSC41-Envelope (E39:X45)

To verify the infiltration values, the four and six side areas were computed:

- Area 4 sides = (11 ft + 9 ft + 9 ft) x (180 + 180 + 60 + 60) = 29 x 480 = 13,920 ft2
- Area 6 sides = Area 4 sides + 180 x 60 x 2 = Area 4 sides + 21,600 = 35,520 ft2
- Ratio of 6 sides to 4 sides = 35,520/13,920 = 2.552

So, according to the referenced report

$$I_{design} \text{ at the building envelope } = 0.112 \times I_{75} \text{ at building envelope} \qquad 2$$

$$I_{design\ (4-side)} = I_{design\ (5-or\ 6-side)} \times \left(\frac{\text{Total leakage area } (5-\text{ or } 6-\text{ side})}{\text{Exterior Wall area}}\right) \qquad \textit{Equation 3}$$

Using these equations, the values should be:

Pre 1978	0.314
1978-1997	0.314
1998-2007	0.314
2008-2018	0.258
2019+	0.171

These do not match the value shown in the infiltration table.

SHSC42-HVAC System (A2:G6)

The HVAC systems selected for the existing vintage prototype seem reasonable as well as the use of PVAV for non-guest room portions of the prototype that are adjacent to one another on the first floor such as the lobby, office, conference room, rest room, gym, and first floor hallway but are less likely to be applied to the hallways on the second and third floors. Those hallways are often conditioned using PTACs just like the rest of the guest rooms on those floors, often with a unit at either end of the hall.

SHSC43-HVAC System (A2:G8)

The use of gas heating for guestrooms in new construction seems unusual. Most three-story highway hotels have simple under-the-window, slide-out, PTAC units that are all electric. They are probably more likely to be PTHP rather than units that require either a gas line for a self-contained furnace (which I'm not sure they even make) or a PTAC with a hydronic coil, which would require piping or tubing to the individual PTAC unit (which I'm also not sure they make). I am aware that the Title 24-2025 non-res ACM manual does specify System1 – RAC for hotel/motel guestrooms. This is described below:

Hotel/motel guestrooms	No limit	All	No limit	System 1 – RAC

System 1 – RAC	Residential air conditioner	Single-phase single-zone system with constant volume fan, no economizer, direct expansion cooling, and gas furnace heating.

Unfortunately, gas furnace heating just does not seem reasonable, and this would more likely be a PTHP.

SHSC44-HVAC System (C6)

Continuing from the last comment, assuming System 1-RAC is being used in the prototype, cell C6 says “Gas Boiler,” and it really should be “Gas Furnace.”

SHSC45-HVAC System (C14:C17)

While it may not matter since the EERs appear to have not changed for this equipment, the lifetime of a PTAC unit is probably not much longer than 15 years (according to ASHRAE 2023 Handbook on HVAC Applications Chapter 38, Table 4):

Table 4 Comparison of Service Life Estimates

Equipment Item	Median Service Life, Years: Abramson et al. (2005)	Median Service Life, Years: Akalin (1978)	Equipment Item	Median Service Life, Years: Abramson et al. (2005)	Median Service Life, Years: Akalin (1978)	Equipment Item	Median Service Life, Years: Abramson et al. (2005)	Median Service Life, Years: Akalin (1978)
Air Conditioners			**Air Terminals**			**Condensers**		
Window unit	N/A*	10	Diffusers, grilles, and registers	N/A*	27	Air-cooled	N/A	20
Residential single or split package	N/A*	15	Induction and fan-coil units	N/A*	20	Evaporative	N/A*	20
Commercial through-the-wall	N/A*	15	VAV and double-duct boxes	N/A*	20	**Insulation**		
Water-cooled package	>24	15	**Air washers**	N/A*	17	Molded	N/A*	20
Heat pumps			**Ductwork**	N/A*	30	Blanket	N/A*	24
Residential air-to-air	N/A*	15	**Dampers**	N/A*	20	**Pumps**		
Commercial air-to-air	N/A*	15	**Fans**	N/A*		Base-mounted	N/A*	20
Commercial water-to-air	>24	19	Centrifugal	N/A*	25	Pipe-mounted	N/A*	10
Roof-top air conditioners			Axial	N/A*	20	Sump and well	N/A*	10
Single-zone	N/A*	15	Propeller	N/A*	15	Condensate	N/A*	15
Multizone	N/A*	15	Ventilating roof-mounted	N/A*	20	**Reciprocating engines**	N/A*	20
Boilers, Hot-Water (Steam)			**Coils**			**Steam turbines**	N/A*	30
Steel water-tube	>22	24	DX, water, or steam	N/A*	20	**Electric motors**	N/A*	18
Steel fire-tube		25	Electric	N/A*	15	**Motor starters**	N/A*	17
Cast iron	N/A*	35	**Heat Exchangers**			**Electric transformers**	N/A*	30
Electric	N/A*	15	Shell-and-tube	N/A*	24	**Controls**		
Burners	N/A*	21	**Reciprocating compressors**	N/A*	20	Pneumatic	N/A*	20
Furnaces			**Packaged Chillers**			Electric	N/A*	16
Gas- or oil-fired	N/A*	18	Reciprocating	N/A*	20	Electronic	N/A*	15
Unit heaters			Centrifugal	>25	23	**Valve actuators**		
Gas or electric	N/A*	13	Absorption	N/A*	23	Hydraulic	N/A*	15
Hot-water or steam	N/A*	20	**Cooling Towers**			Pneumatic	N/A*	20
Radiant heaters			Galvanized metal	>22	20	Self-contained		10
Electric	N/A*	10	Wood	N/A*	20			
Hot-water or steam	N/A*	25	Ceramic	N/A*	34			

*N/A: Not enough data yet in Abramson et al. (2005). Note that data from Akalin (1978) for these categories may be outdated and not statistically relevant. Use these data with caution until enough updated data are accumulated in Abramson et al.

But this is probably a bit optimistic according to other sources, which cite seven years on average and 10 years when maintained properly:

https://www.greenlodgingnews.com/extend-ptac-life-improve-efficiency-through-preventive-maintenance/

SHSC46-HVAC System (C14:C18)

The DX Cooling-Guest Rooms number does not seem consistent with requirements:

TABLE 7 TO § 431.97(c)—MINIMUM EFFICIENCY STANDARDS FOR PTAC AND PTHP

Equipment type	Category	Cooling capacity	Efficiency level	Compliance date: products manufactured on and after . . .
PTAC	Standard Size	<7,000 Btu/h	EER = 11.7	October 8, 2012.[2]
		≥7,000 Btu/h and ≤15,000 Btu/h	EER = 13.8 − (0.3 × Cap [1])	October 8, 2012.[2]
		>15,000 Btu/h	EER = 9.3	October 8, 2012.[2]
	Non-Standard Size.	<7,000 Btu/h	EER = 9.4	October 7, 2010.
		≥7,000 Btu/h and ≤15,000 Btu/h	EER = 10.9 − (0.213 × Cap [1])	October 7, 2010.
		>15,000 Btu/h	EER = 7.7	October 7, 2010.
PTHP	Standard Size	<7,000 Btu/h	EER = 11.9 COP = 3.3	October 8, 2012.
		≥7,000 Btu/h and ≤15,000 Btu/h	EER = 14.0 − (0.3 × Cap [1]) COP = 3.7 − (0.052 × Cap [1])	October 8, 2012.
		>15,000 Btu/h	EER = 9.5 COP = 2.9	October 8, 2012.
	Non-Standard Size.	<7,000 Btu/h	EER = 9.3 COP = 2.7	October 7, 2010.
		≥7,000 Btu/h and ≤15,000 Btu/h	EER = 10.8 − (0.213 × Cap [1]) COP = 2.9 − (0.026 × Cap [1])	October 7, 2010.
		>15,000 Btu/h	EER = 7.6 COP = 2.5	October 7, 2010.

[1] "Cap" means cooling capacity in thousand Btu/h at 95 °F outdoor dry-bulb temperature.
[2] And manufactured before January 1, 2017. See Table 8 of this section for updated efficiency standards that apply to this category of equipment manufactured on and after January 1, 2017.

The EER=9 shown in the cells C14:C18 for a 1-ton PTAC, the standard size unit would result in:

$$EER = 13.8 - 0.3 \times (12{,}000/1000) = 10.2$$

A 1.3-ton PTAC would result in an EER of 9.0, but that is large for what is needed for a typical hotel guest room. Please explain more about the details of how the EER of 9 was developed.

SHSC47-HVAC System (D18)

For DX Cooling-All Other Conditioned Zones for new construction, the value shown in cell D18 is: "EER=10 (≥ 240,000 Btu/h and < 760,000 Btu/h) and EER=9 (<65,000 Btu/h)" and references CFR Energy 2025, Table 12. Based on Table 3 below, the larger units should be IEER 13.2 for electric resistance heating and IEER 13.0 for other heating. For smaller units, Table 19 shows SEER2 13.4. These would need to be translated into the proper inputs for EnergyPlus.

TABLE 3 TO § 431.97(b)—UPDATES TO THE MINIMUM COOLING EFFICIENCY STANDARDS FOR AIR CONDITIONING AND HEATING EQUIPMENT

[Not including single package vertical air conditioners and single package vertical heat pumps, packaged terminal air conditioners and packaged terminal heat pumps, computer room air conditioners, variable refrigerant flow multi-split air conditioners and heat pumps, and double-duct air-cooled commercial package air conditioning and heating equipment]

Equipment type	Cooling capacity	Subcategory	Heating type	Efficiency level	Compliance date: equipment manufactured starting on . . .
Small Commercial Packaged Air Conditioning and Heating Equipment (Air-Cooled).	≥65,000 Btu/h and <135,000 Btu/h.	AC	Electric Resistance Heating or No Heating.	IEER = 12.9	January 1, 2018.[1]
				IEER = 14.8	January 1, 2023.
			All Other Types of Heating	IEER = 12.7	January 1, 2018.[1]
				IEER = 14.6	January 1, 2023.
		HP	Electric Resistance Heating or No Heating.	IEER = 12.2	January 1, 2018.[1]
				IEER = 14.1	January 1. 2023.
			All Other Types of Heating	IEER = 12.0	January 1, 2018.[1]
				IEER = 13.9	January 1, 2023.
Large Commercial Packaged Air Conditioning and Heating Equipment (Air-Cooled).	≥135,000 Btu/h and <240,000 Btu/h.	AC	Electric Resistance Heating or No Heating.	IEER = 12.4	January 1, 2018.[1]
				IEER = 14.2	January 1, 2023.
			All Other Types of Heating	IEER = 12.2	January 1, 2018.[1]
				IEER = 14.0	January 1, 2023.
		HP	Electric Resistance Heating or No Heating.	IEER = 11.6	January 1, 2018.[1]
				IEER = 13.5	January 1, 2023.
			All Other Types of Heating	IEER = 11.4	January 1, 2018.[1]
				IEER = 13.3	January 1, 2023.
Very Large Commercial Packaged Air Conditioning and Heating Equipment (Air-Cooled).	≥240,000 Btu/h and <760,000 Btu/h.	AC	Electric Resistance Heating or No Heating.	IEER = 11.6	January 1, 2018.[1]
				IEER = 13.2	January 1, 2023.
			All Other Types of Heating	IEER = 11.4	January 1, 2018.[1]
				IEER = 13.0	January 1, 2023.
		HP	Electric Resistance Heating or No Heating.	IEER = 10.6	January 1, 2018.[1]
				IEER = 12.5	January 1, 2023.
			All Other Types of Heating	IEER = 10.4	January 1, 2018.[1]
				IEER = 12.3	January 1, 2023.
Small Commercial Packaged Air-Conditioning and Heating Equipment (Water-Source: Water-to-Air, Water-Loop).	<17,000 Btu/h	HP	All	EER = 12.2	October 9, 2015.
	≥17,000 Btu/h and <65,000 Btu/h.	HP	All	EER = 13.0	October 9, 2015.
	≥65,000 Btu/h and <135,000 Btu/h.	HP	All	EER = 13.0	October 9, 2015.

[1] And manufactured before January 1, 2023.

TABLE 19 TO § 431.97(h)—UPDATED MINIMUM EFFICIENCY STANDARDS FOR AIR-COOLED, THREE-PHASE, SMALL COMMERCIAL PACKAGE AIR CONDITIONING AND HEATING EQUIPMENT WITH A COOLING CAPACITY OF LESS THAN 65,000 Btu/h AND AIR-COOLED, THREE-PHASE, SMALL VARIABLE REFRIGERANT FLOW MULTI-SPLIT AIR CONDITIONING AND HEATING EQUIPMENT WITH A COOLING CAPACITY OF LESS THAN 65,000 Btu/h

Equipment type	Size category (cooling)	Subcategory	Minimum efficiency	Compliance date: equipment manufactured starting on . . .
Air Conditioners	<65,000 Btu/h.	Split-System	13.4 SEER2	January 1, 2025.
		Single-Package.	13.4 SEER2	January 1, 2025.
Heat Pumps	<65,000 Btu/h.	Split-System	14.3 SEER2, 7.5 HSPF2.	January 1, 2025.
		Single-Package.	13.4 SEER2, 6.7 HSPF2.	January 1, 2025.
Space-Constrained Air Conditioners.	≤30,000 Btu/h.	Split-System	12.7 SEER2	January 1, 2025.
		Single-Package.	13.9 SEER2	January 1, 2025.
Space-Constrained Heat Pumps.	≤30,000 Btu/h.	Split-System	13.9 SEER2, 7.0 HSPF2.	January 1, 2025.
		Single-Package.	13.9 SEER2, 6.7 HSPF2.	January 1, 2025.
Small-Duct, High-Velocity Air Conditioners.	<65,000 Btu/h.	Split-System	13.0 SEER2	January 1, 2025.
Small-Duct, High-Velocity Heat Pumps.	<65,000 Btu/h.	Split-System	14.0 SEER2, 6.9 HSPF2.	January 1, 2025.
VRF Air Conditioners	<65,000 Btu/h.	.....	13.4 SEER2	January 1, 2025.
VRF Heat Pumps	<65,000 Btu/h.	.....	13.4 SEER2, 7.5 HSPF2.	January 1, 2025.

SHSC48-HVAC System (A22:H31)

Confirming fan power inputs with the PNL 20405 report

- Existing PTAC/PTHP TSP 1.33 – ok

Confirming fan power inputs with ACM 2025

- Existing PTAC/PTHP Fan Efficiency 0.65- ok
- New construction Fan Efficiency 0.65 - ok
- New construction System 3b-SZHP, <5000 CFM 0.744 – ok
- New construction System 5-PVAV, >10000 CFM 0.964 – ok
- New construction fan motor and drive efficiency – needs addressing (see comment SHSC50-HVAC System)

SHSC49-HVAC System (D23:D31)

The header indicates that the source for the Total Static Pressure is PNNL-20405, but that is only used for the existing vintage rows. The other rows are based on calculations using the fan formula and inputs from the ACM. Please correct the header. This is the same comment as LHSC51-HVAC System.

SHSC50-HVAC System (F28:F31)

For new construction, a fixed value of 0.9 for the fan motor and drive efficiency is not correct, and instead, the size of the motor and ACM 2025 Table 13 should be used. This is described as the process for existing vintages but not for new construction. This is the same comment as LHSC52-HVAC System.

SHSC51-Thermostat (A7:R10)

The corridor heating (68°F constant) and cooling (78°F constant) setpoints for corridors are based on the Appendix 5.4B spreadsheet (the version I referenced was named “CEC-400-2022-009-CMF-REV2-AP5.4B.xlsx”) and the “ResidentialCommon” tab. Please add the tab name to the reference. This is the same comment as LHSC54-Thermostat.

SHSC52-Thermostat (A3:R6)

The non-res common space heating (70°F with 60°F setback) and cooling (75°F with 85°F) setpoints seem to be based on the Appendix 5.4B spreadsheet (the version I referenced was named “CEC-400-2022-009-CMF-REV2-AP5.4B.xlsx”) and the “Assembly” tab. Please add the tab name to the reference. This is the same comment as LHSC55-Thermostat.

SHSC53-Thermostat (A11:R14)

For guestrooms, the thermostat setting is a complicated topic. Using a 70°F constant setpoint for heating and 72°F based on a default is a simple approach, but I would recommend a more comprehensive approach for the prototype. Some considerations to include would be:

- Diversity of temperatures that hotel guests find comfortable
- Hotel guests do not pay extra for energy use to obtain greater comfort
- Oversetting of thermostats upon initial entry of the room (exaggeration of setpoint need)
- Thermostats with occupancy sensors and their impact on sleeping guests
- BAS resets setpoints when no guest is present
- Hotel staff resetting setpoints

To try to capture these effects, use several different schedules for guest rooms that have a wide diversity of values, including deep setbacks for unoccupied rooms. This is the same comment as LHSC56-Thermostat.

SHSC54-Ventilation (B1:B50)

The number of occupants in different zones directly impacts the outdoor airflow rates, and comments on the number of occupants were already covered in comments: SHSC21-Zones (office size), SHSC27-Zones (lobby), and SHSC28-Zones (guest rooms). Specifically, the guest room ventilation needs to be based on maximum expected occupancy, which is generally four (since hotels often switch rooms that can have two double or queen beds with rooms with a single queen or king bed). This is very similar to the comment as LHSC57-Ventilation.

SHSC55-Ventilation (A1:K50)

No differentiation by vintage was included for outdoor airflow. Earlier vintages of hotels probably had significantly different ventilation requirements. Since this often impacts the sizing of ducts, intake vents, and air-handling equipment, it is unlikely to be upgraded very often. This is the same comment as LHSC58-Ventilation.

SHSC56-Ventilation (F44:F49)

Exhaust air flow is shown as required for stairways and storage rooms, shown as 50/70 cfm/unit. Stairways and storage rooms should have no exhaust. I believe this was pasted by accident in these cells.

SHSC57-Ventilation (F16:F40)

The required exhaust air flow for guestrooms is shown as 20/50 cfm/unit per shower head. This is based on Table 120.1-B in Title 24-2025, but two different rows may apply to the bathroom in each guestroom:

Table 120.1-B – Minimum Exhaust Rates [ASHRAE 62.1: Table 6-2]

Occupancy Category	Exhaust Rate, cfm/unit	Exhaust Rate, cfm/ft^2	Air Class	Notes
Shower rooms	20/50	-	2	G,H
Paint spray booths	-	-	4	F
Parking garages	-	0.75	2	C
Pet shops (animal areas)	-	0.90	2	-
Refrigerating machinery rooms	-	-	3	F
Soiled laundry storage rooms	-	1.00	3	F
Storage rooms, chemical	-	1.50	4	F
Toilets – private	25/50	-	2	E

The value of 25/50 cfm may make more sense for each guestroom. Note that the two values are for continuous and intermittent fan operation. The first value, 25 CFM, should be used for most hotel applications. This value ends up being smaller than the CFM driven by flow rates, but exhaust flow should be documented here.

SHSC58-Water Heater (B10:B57, E10:E57)

The peak water heating consumption table columns for “Baseline WH Type” and “Hot Water Usage Specification” were verified against the source T24, 2022, Appendix 5.4A,

sheet: “2022 SpaceFuncData-Input”, and all were ok. It would be helpful to indicate the NACM Space Type assumed for each zone so that they could be more easily checked.

SHSC59-Water Heater (A4:B5, A10:B57)

The use of both electric and gas water heating equipment for new construction in the same building is quite unusual, except in the case of small electric water heaters being used under remote washrooms for sinks. This is unlikely in a hotel that has extensive piping to serve all the guest rooms already. Based on this, all water heating should probably be gas. This is the same comment as LHSC61-Water Heater.

SHSC60-Water Heater (A66:B66)

The design supply temperature is shown as 135°F, but the source for this Appendix 5.4B (called CEC-400-2022-009-CMF-REV2-AP5.4B.xlsx) on the “ResidentialLiving” tab shows a constant 130°F as the “WtrHtrSetpt.” This is the same comment as LHSC63-Water Heater.

SHSC61-Water Heater (A67:B67)

The cold water supply temperature of 55°F shown in the scorecard is consistent with the citation of the ACM 5.9.2 Rated Capacity, but better data is available that may be closer to how water heaters are actually sized using the local water mains temperature. The Appendix 5.4B (called CEC-400-2022-009-CMF-REV2-AP5.4B.xlsx) includes a WaterMain tab and WaterMainMonth tab that show climate-specific values for the entering water temperature. The lowest of these values for each climate zone is more likely to be used for sizing water heaters; the monthly values vary from 41.4 in CZ16 to 63.2 in CZ15. Using these values would have a large impact on the sizing of the water heater. This is the same comment as LHSC64-Water Heater.

SHSC62-Interior Lights (B4:B14)

LPDs are consistent with the cited source for new construction. The lobby+lounge used 0.55 W/ft2 in this prototype, but the large hotel used 0.70 W/ft2 for the lobby. They should probably be made consistent.

SHSC63-Interior Lights (E4:E14)

LPDs are consistent with the cited source for existing buildings, except that the LPD for the guest room value of 1.27 was not checked.

SHSC64-Interior Lights (E10)

The lighting power density for the guest room, with a value of 1.27, seems very high. A value of 0.41 is suggested in the notes and would be consistent with the 90.1-2019 version value. This is the same comment as LHSC69-Interior Lights.

SHSC65-Interior Lights (E4:E14)

The use of T24-2019 for existing buildings is based on the comment:

“Hotels are expected to update their lighting systems more frequently than offices. Therefore, the code edition of 2019 was considered as the reference for LPD of the existing prototypes.”

This seems a bit optimistic for how often a hotel does a lighting refresh. Using Title 24-2013 might be more realistic. Of course, the same type of comment applies as stated for other system-related replacements. It might be better to average several different versions of Title 24 together rather than assume that all existing buildings have updated their systems during a specific time. In this case, I would suggest that versions of Title 24 from 2008, 2010, 2013, 2016, and 2019 all be examined, and an engineering estimate of the weight of each be used to determine the LPD for each space. This is the same comment as LHSC70-Interior Lights.

SHSC66-Ext Lights (E9:E15)

The lighting power allowances from Title 24-2025 were checked and are ok. The assumption that lighting zone 2 is being used should be included.

SHSC67-Ext Lights (n/a)

Where are the exterior lighting assumptions for the vintage models? One assumes that those should be higher. This is the same comment as LHSC72-Ext Lights.

SHSC68-Ext Lights (B9:C15)

Please document the sources used for assumptions for computed areas and counts. It mentions the 2019 Outdoor Lighting Case Report and PNNL 90.1-2010 TSD, but those are both big, complicated documents. More specific references would enable the values to be checked. This is the same comment as LHSC73-Ext Lights.

SHSC69-Equipment (A6:D6, A9:D9)

The reference "Plug load and schedules project with CEC, 2025" could not be found to verify many of the numbers shown on the table. This is the same comment as LHSC74-Equipment.

SHSC70-Equipment (B2:C13)

The Plug Load and Gas Power Density for all numbers using Title 24-2022 Appendix 5.4A were all verified using the spreadsheet titled "CEC-400-2022-009-CMF-REV2-AP5.4A_0.xlsx" on the "2022 SpaceFuncData-Input" tab, except the "Lobby+Lounge" space. The lobby+lounge used 1.0 W/ft2 in this prototype, but the large hotel used 0.50 W/ft2 for the lobby. They should probably be made consistent since they are presumably the same functional space.

SHSC71-Equipment (B17:B18)

The number of elevators assumed for the prototype of 2 seems reasonable.

SHSC72-Equipment (A206:C207)

The elevator-related notes should be removed:

- One elevator for approximately 50,000 ft^2 in use.
- For every two floors or two and a half floors, one more elevator was added.

These notes correspond to an article that was cited in the large hotel scorecard but is not present here. The 50,000 was related to office buildings, and the rule of thumb for adding more is not necessary for this building. This is related to LHSC76-Equipment.

SHSC73-Equipment (D18)

The lighting shutoff for elevators should be after 15 minutes according to 120.6 (f)(3):

> 3. When the elevator cab is stopped and unoccupied with doors closed for over 15 minutes, the cab interior lighting and ventilation fans shall be switched off until elevator cab operation resumes.

Not 5 minutes as shown.

SHSC74-Equipment (H18, A211:C211)

The design load for the new construction elevator is shown as 9 kW but should be based on the ACM 2025 requirement of 10 kW, shown in 5.4.8:

> *Input Restrictions:* The power values are prescribed as 10 kW per elevator and 3.93 kW per escalator or moving walkway for the proposed design.

And the reference to the power in A211 should reference ACM 2025 and the section number 5.4.8.

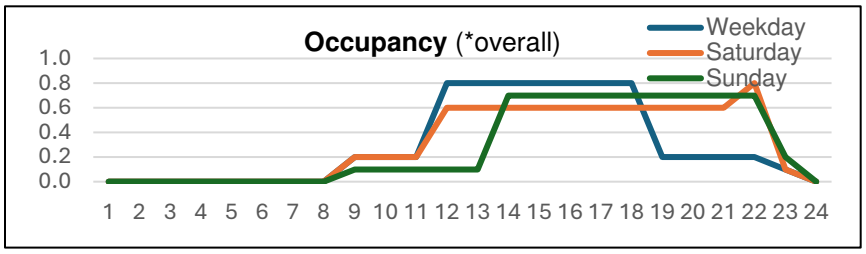
SHSC75-Schedules (AC47:AI49, AC66:AI71)

Several of the graphs shown on this tab are incorrectly labeled, including:

- Corridor infiltration (shown as elevator)
- Guest rooms elevator (shown as SHW)
- Guest rooms infiltration (shown as SHW)

SHSC76-Schedules (A4:A6)

The “Assembly Occupancy” is used for “All spaces except Guest rooms and Corridors”



And based on this statement, it applies to zones:

- Lobby+Lounge
- Gym
- Closed office
- Storage
- Laundry rooms, central
- Mech/Elec
- Stairs
- RestRoom

For many of these spaces, the schedule does not work well:

- Lobby+Lounge – the 80% occupancy assumption from noon to 6 pm, coupled with the 70 occupants in the hotel with only 56 rooms, seems unlikely. It is more likely to peak in the morning and in the evening when most travelers are arriving and leaving a hotel. In addition, the lobby of most highway hotels of this size often has some type of continental breakfast, which usually makes the busiest times in the lobby between 7 am and 9 am. Check-in time usually begins at 3 pm or 4 pm, and most arrivals are typically after that time until 8 pm. The weekday and weekend schedules should be made more consistent in shape, with Friday, Saturday, and Sunday being higher than other days in the week due to more families traveling on weekends. While this assessment is anecdotal, it is probably better than the assumed schedule for the lobby+lounge. Also the HVAC availability schedule for the lobby should be 24x7.
- Gym – the peak time for a gym in a hotel is the morning, often early from 6 am and lasting until 9 am. Some additional activity in the evening is common, but not as busy as the morning.
- Closed office – presuming this is the hotel desk staff and management, these offices are somewhat occupied 24 hours a day, although the desk staff is usually encouraged to be at the front desk.
- Storage – no comment
- Laundry rooms, central – the laundry room in a hotel is driven primarily by the need of housekeeping staff and the changing of linens, and should peak in the late morning and stay busy until early evening, and is probably consistent.
- Mech/Elec – no comment
- Stairs – no comment
- RestRoom – it should follow the schedule of the lobby+lounge

The lighting, plug-load, HVAC availability, and SWH scheduled should be modified for these spaces to be consistent with any revised occupancy schedules.

SHSC77-Schedules (A41:AI43)

There is no corridor service hot water usage according to the zone tab, so this schedule is not necessary. This is the same comment as LHSC85-Schedules.

SHSC78-Schedules (A13:AI15)

The lobby HVAC availability schedule should probably be 24x7, and the infiltration for the lobby should match that. Depending on how open the retail and dining areas on the first floor are (they often share a large open space with the lobby), they may need to have the same schedule. This is the same comment as LHSC84-Schedules.

SHSC79-Schedules (x:x)

The guest room occupancy schedule (and most others) is not different by the days of the week. I believe most hotels are significantly busier on Friday and Saturday evenings due to greater family travel.

SHSC80-Schedules (A51:AI53)

The occupancy schedules for different hotel rooms should be different to represent the diversity of arrival and departure times and the number of occupants (see also SHSC28-Zones). It is interesting to note that the SHW schedule shows earlier peak use of hot water during weekdays than on weekends. This makes sense and should be reflected in the occupancy schedule.

SHSC81-Schedules (A19:AI21, A44:AI46, A66:AI68)

The elevator schedule is shown in the Assembly Elevator, Corridor Elevator, and the Guest Rooms Elevator. While the Assembly Elevator, Corridor Elevator, and the Guest rooms Elevator are the same, I presume the elevator is associated with the corridor and not the guest rooms. The Assembly Elevator is quite different, but I don't think it is being used. Only one elevator schedule that is actually being used in the prototypes should be included.

SHSC82-Schedules (Column B)

Shouldn’t the schedules be based on Title 24-2025 instead of Title 24-2022?

SHSC83-Schedules (A1:AB71)

All schedule values were verified against the Title 24-2022 file “CEC-400-2022-009-CMF-REV2-AP5.4B.xlsx” and were correct. Note that the Guest room equipment schedule is shown as from “Plug load and schedules project with CEC, 2025” and was not checked.